QUESTION 1

What is the main difference between a sequential element and a combinational element in digital circuits?

Answer is Correct **Score: 1 / 1**

Options

A. Sequential elements depend only on current inputs

Correct

B. Sequential elements store state and depend on clock transitions

C. Combinational elements store state and depend on previous inputs

D. Sequential elements change output instantly with input changes

FEEDBACK

Solution: Sequential elements retain state and their output depends on both previous inputs and the timing of clock transitions.

Link: https://youtu.be/yjVw0hwQsBc?t=279

QUESTION 2

Why are edge-triggered flip-flops more commonly used than other sequential elements?

Answer is Correct **Score: 1 / 1**

Options

A. They require fewer components

Correct

B. They are easier to analyze for critical paths and delays

C. They operate without a clock signal

D. They consume less power

FEEDBACK

Solution: Edge-triggered flip-flops simplify the analysis of circuit timing and critical paths, making them popular in digital systems.

Link: https://youtu.be/yjVw0hwQsBc?t=374

QUESTION 3

What does the positive edge of a clock signal represent?

Answer is Correct **Score: 1 / 1**

Options

A. Clock transition from high to low

Correct

B. Clock transition from low to high

C. A constant high clock level

D. A constant low clock level

FEEDBACK

Solution: A positive edge occurs when the clock transitions from 0 to 1

QUESTION 4

What is the primary reason for not indefinitely increasing the clock frequency of a processor?

Answer is Correct **Score: 1 / 1**

Options

A. Software limitations

B. Human reaction time limits

Correct

C. Physical gate delays due to electrical characteristics

D. Lack of available crystals

FEEDBACK

Solution: The speed is limited by the finite delays of transistors, resistors, and capacitors that make up logic gates.

Link: https://youtu.be/yjVw0hwQsBc?t=664

QUESTION 5

According to the dynamic power equation, which factor affects power consumption quadratically?

Answer is Correct **Score: 1 / 1**

Options

A. Capacitance (C)

Correct

B. Voltage (V)

C. Frequency (f)

D. Number of gates

FEEDBACK

Solution: Power consumption is proportional to the square of the supply voltage.

Link: https://youtu.be/yjVw0hwQsBc?t=1265

QUESTION 6

Which method can reduce power consumption without changing voltage or capacitance?

Answer is Correct **Score: 1 / 1**

Options

A. Increasing the clock speed

B. Using a larger circuit

Correct

C. Reducing operating frequency

D. Using edge-triggered flip-flops

FEEDBACK

Solution: Lowering the operating frequency reduces the switching activity and thus power consumption.

Link: https://youtu.be/yjVw0hwQsBc?t=1351

QUESTION 7

Why do microcontrollers often have multiple clocks?

Not Attempted **Score: 0 / 1**

Options

A. To avoid using an external oscillator

Expected

B. To allow different parts of the system to run at different speeds for performance-power trade-offs

C. To simplify programming

D. To avoid temperature variations

FEEDBACK

Solution: Multiple clocks allow selective control of peripheral speeds, improving power efficiency.

Link: https://youtu.be/yjVw0hwQsBc?t=1461

QUESTION 8

What is a common frequency for a real-time clock (RTC) crystal, and why?

Not Attempted **Score: 0 / 1**

Options

A. 1 MHz – easy to divide into microseconds

B. 48 MHz – matches system clock

Expected

C. 32.768 kHz – divides exactly to 1 Hz

D. 100 kHz – stable for long periods

FEEDBACK

Solution: 32,768 Hz is 2^15, allowing exact division to get a 1-second tick.

Link: https://youtu.be/yjVw0hwQsBc?t=2340

QUESTION 9

Which oscillator type is built into most microcontrollers but is less accurate?

Not Attempted **Score: 0 / 1**

Options

A. Crystal oscillator

Expected

B. RC oscillator

C. PLL

D. Clock divider

FEEDBACK

Solution: RC oscillators are built-in but can have errors up to several percent, making them less accurate than crystals.

Link: https://youtu.be/yjVw0hwQsBc?t=1924

QUESTION 10

What is the main function of a Phase Locked Loop (PLL) in a microcontroller clock system?

Not Attempted **Score: 0 / 1**

Options

A. Reduce voltage fluctuations

B. Generate a stable 32 kHz signal

Expected

C. Convert a lower frequency to a higher frequency

D. Store clock cycles

FEEDBACK

Solution: PLLs are used to multiply the input clock frequency to higher values.

Link: https://youtu.be/yjVw0hwQsBc?t=2080

QUESTION 11

What is the purpose of prescalers in a microcontroller clock system?

Not Attempted **Score: 0 / 1**

Options

A. Increase clock frequency

Expected

B. Divide a master clock into lower frequencies

C. Store timing information

D. Synchronize with external devices

FEEDBACK

Solution: Prescalers reduce a high-frequency clock to a desired lower frequency.

Link: https://youtu.be/yjVw0hwQsBc?t=2432

QUESTION 12

What does “dynamic frequency scaling” allow in microcontrollers?

Not Attempted **Score: 0 / 1**

Options

A. Permanently setting the system to the highest speed

Expected

B. Changing the clock frequency at runtime

C. Synchronizing multiple microcontrollers

D. Disabling unused peripherals

FEEDBACK

Solution: Dynamic frequency scaling lets the system adjust its clock frequency during operation to balance performance and power.

Link: https://youtu.be/yjVw0hwQsBc?t=2432